

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U30-1



S23-3445U30-1

TUESDAY, 16 MAY 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

FOUNDATION TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	11	
3.	12	
4.	10	
5.	13	
6.	5	
7.	14	
Total	75	

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01

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator, a pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(a) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer is incorrect.

The Periodic Table is printed on the back cover of the examination paper.



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Answer **all** questions in the spaces provided.

1. Ash dieback disease is a fungal infection that is common in ash trees in Wales.

The disease causes the leaves to blacken and wilt, so eventually the ash tree has no leaves.

The photographs show healthy ash leaves and those affected by ash dieback.



- (a) Photosynthesis occurs in green leaves.

- (i) Use word(s) from the box to complete the paragraph below. You may use each word once, more than once or not at all. [5]

stomata	nitrogen	chlorophyll	oxygen	glucose
	water	carbon dioxide		

Leaves use to absorb light energy.

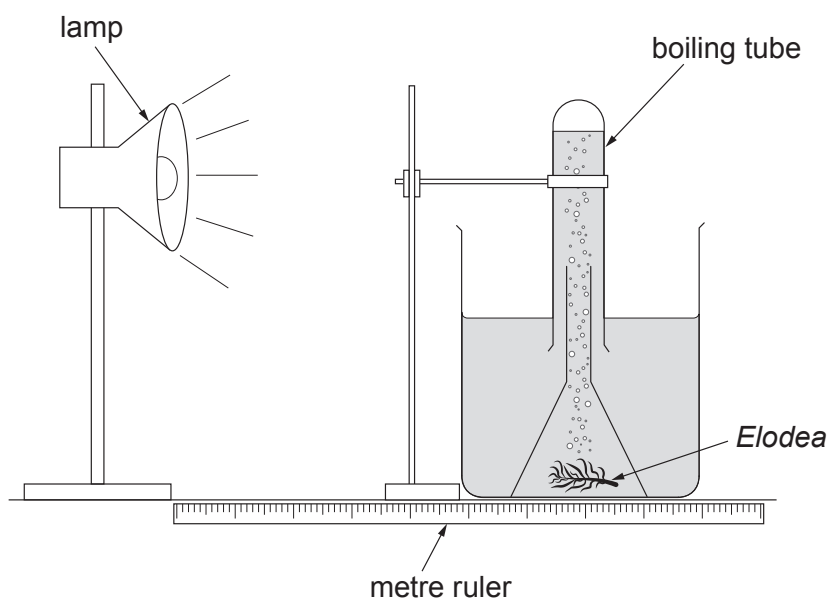
They convert and into
..... and

- (ii) Complete the following sentence by underlining the correct word in the brackets. [1]

Ash dieback disease affects the growth of an ash tree as the leaves are unable to make (**food** / **water** / **nitrogen**).



- (b) The following apparatus is used to investigate how light intensity affects the rate of photosynthesis of an *Elodea* plant.



The method used is described below.

1. Add **one spatula** of sodium hydrogen carbonate to a beaker containing 200 cm³ of water at 20°C, to supply carbon dioxide.
2. Place the *Elodea* in the beaker.
3. Completely fill a boiling tube with water and carefully place over the end of the funnel with the end under the water, clamp into place.
4. Place the lamp 5 cm away from the *Elodea*.
5. Start the stopwatch and record the **number of bubbles** of oxygen produced in a **few seconds**.
6. Repeat the experiment with the lamp 10 cm, 15 cm, 20 cm, 25 cm and 30 cm from the apparatus.

- (i) Steps 1 and 5 could lead to uncertainties. In the table below, circle **one** improvement in each row that you would use.

[3]

Current method	Improvement	
add one spatula	add 1 kg	add 1 g
record the number of bubbles	collect the gas in a beaker	collect the gas in a measuring cylinder
produced in a few seconds	produced in 1 minute	produced in 1 hour

- (ii) State how you would change step 6 to investigate the effect of temperature on photosynthesis.

[1]

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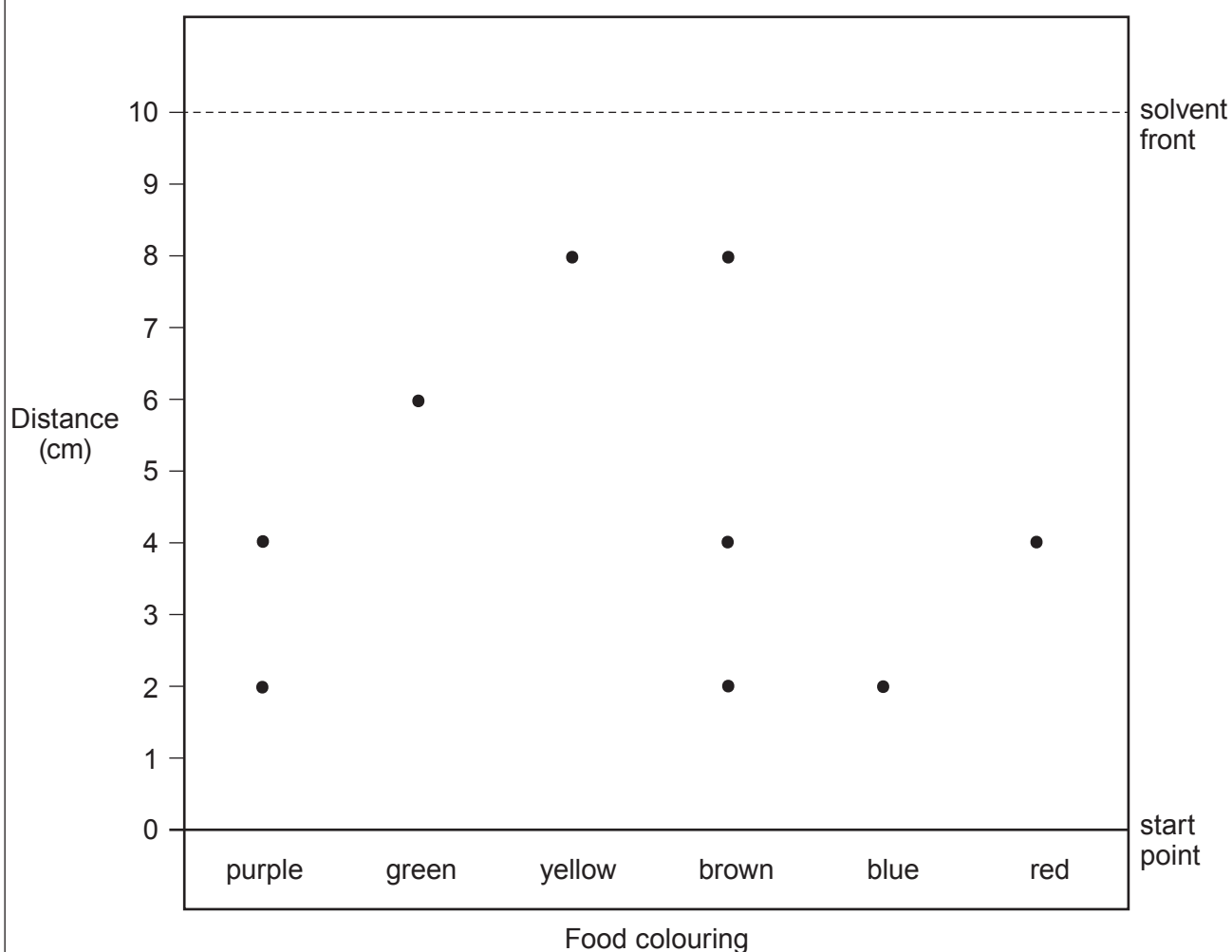
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2. Analytical scientists work in a wide range of different industries and agencies.

- (a) The Food Standards Agency (FSA) funded a project to analyse the colours used in food and drinks.

The diagram below shows a chromatogram of the dyes present in different food colourings.



- (i) It was thought that the dye in the blue colouring would not be present in other colourings.
Explain whether you agree.

[2]

.....

.....



(ii) Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the R_f value for the dye in the **red** colouring.

[3]

$R_f =$

- (b) Analytical scientists also identify unknown compounds.
There were three bottles, each containing a solution of one of the following compounds.

copper(II) sulfate

lithium carbonate

sodium chloride

- (i) State which of these compounds would produce a yellow flame in a flame test. [1]

- (ii) I. State which of these compounds would produce bubbles of gas when added to hydrochloric acid. [1]

- II. The gas turns limewater milky. Name this gas. [1]

- (iii) 1 cm³ of silver nitrate solution is added to three different test tubes, each containing a solution of one of the compounds in the box above.

- I. State which of these compounds will give a white precipitate. [1]

- II. Complete the risk assessment below for silver nitrate. [2]

Hazard	Risk	Control measure
silver nitrate is irritant	could cause eye damage if splashes in the eyes when	
		
		
		



3. Hunterston B is a nuclear power station in Scotland.

(a) The reaction in the reactor is shown in the equation below.



(i) **Complete** the equation. [2]

(ii) Circle the part of the reactor that slows down neutrons so they can be absorbed by uranium nuclei. [1]

moderator fuel rods concrete

(iii) State how many neutrons are produced by the fission of each uranium nucleus. [1]

.....

(b) Nuclear reactor 3 at the Hunterston B power station was shut down after cracks were found in the graphite bricks. This made it difficult for the control rods to move up and down.

(i) Explain the role of control rods. [2]

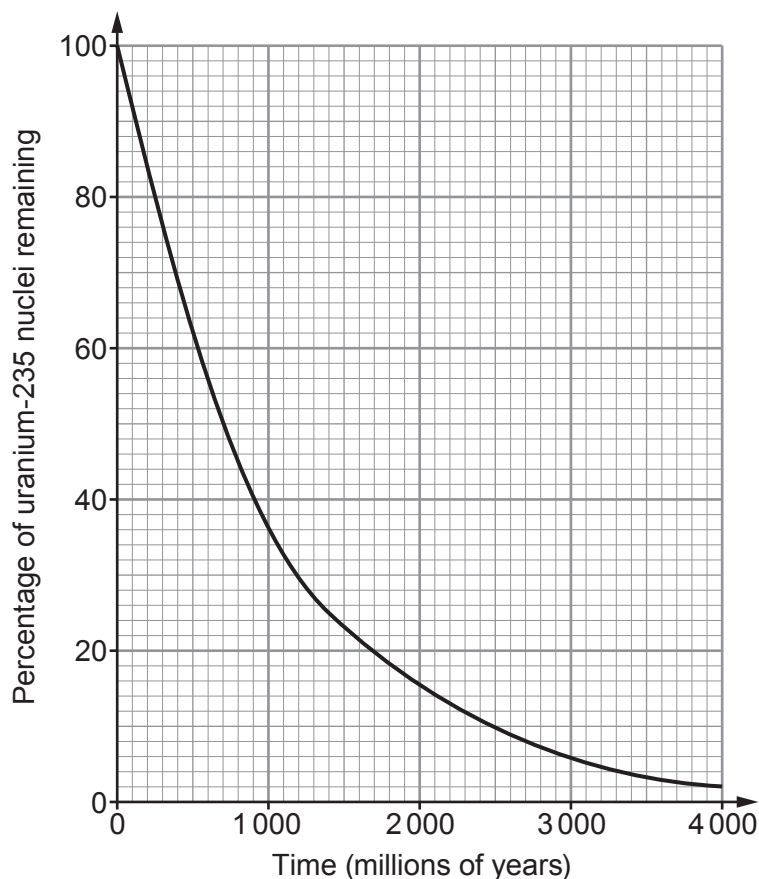
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(ii) An unusual event, such as an earthquake, might move the damaged graphite bricks so that the control rods are unable to move. Explain why this is dangerous. [2]

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- (c) After Hunterston B power station is decommissioned, the uranium-235 in the fuel rods will continue to decay at the rate shown in the graph below.



- (i) Use the graph to answer the following questions.

I. Find the time taken for the uranium-235 to decay from 100% to 50%. [1]

time = million years

II. State the half-life of uranium-235. [1]

half-life = million years

- (ii) The table shows the percentage of uranium-235 nuclei remaining after different half lives. Complete the table. [2]

Number of half-lives	0	1	2	3	4
Percentage of uranium-235 nuclei remaining	100		25		6.25



4. Heating is a common method used to preserve food.

- (a) **State three** methods, **other than heating**, in which the growth of bacteria can be slowed down or stopped in food products.

Explain how each method works.

[6 QER]

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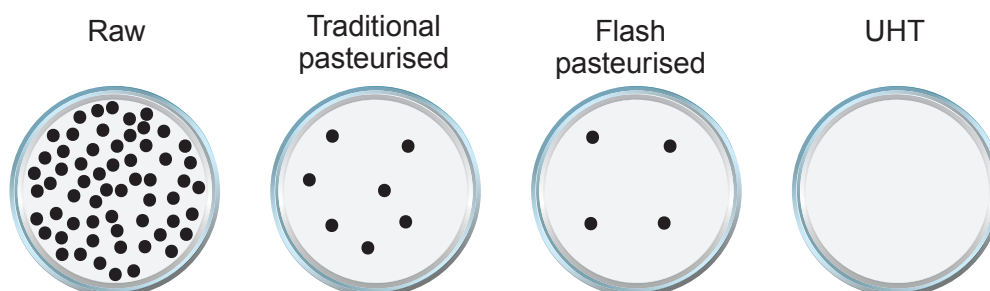
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- (b) Scientists investigated four types of milk. They monitored the growth of bacterial colonies on agar plates using 0.1 cm^3 from each type of milk over two days. Their results are shown below.



Type of milk	Heat treatment	Number of bacterial colonies from 0.1 cm^3 of milk	Number of bacterial colonies in a 40 cm^3 serving
raw	none	60	24 000
traditional pasteurised	63°C for 30 minutes	7	
flash pasteurised	78°C for 35 seconds		1 600
ultra-high temperature pasteurised (UHT)	135°C for 2 seconds	none	none

- (i) Complete the table.

[2]

Space for working

- (ii) It is claimed that unopened UHT milk can be stored on supermarket shelves at room temperature for six months. Explain whether you agree with the claim. [2]

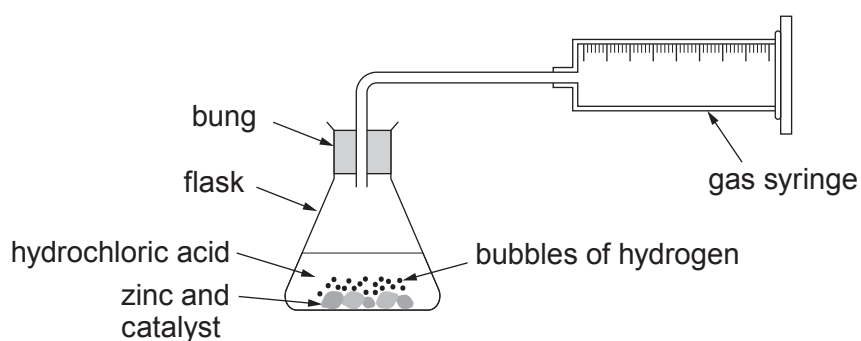
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5. The diagram below shows apparatus that can be used to investigate the rate of the reaction between large zinc pieces and hydrochloric acid. A catalyst was also added to the flask.

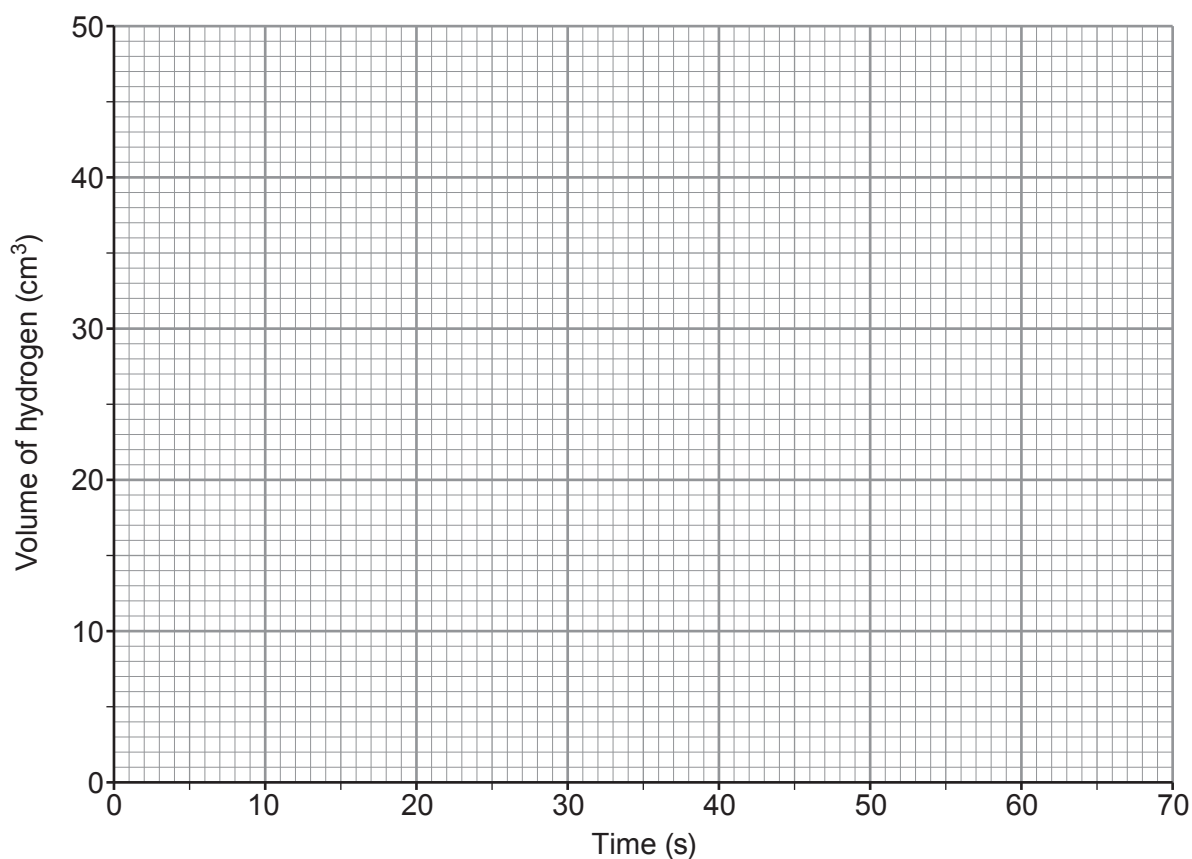


Large pieces of zinc were placed in excess dilute hydrochloric acid and the volume of hydrogen produced was recorded every 15 seconds. The results obtained are shown below.

Time (s)	0	15	30	45	60
Volume of hydrogen (cm ³)	0	28	40	48	48

- (a) (i) Plot the results on the grid below and draw a suitable curve.

[3]



(ii) Use your graph to answer the following questions.

I. Describe the relationship between volume of hydrogen produced and time.

[2]

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.....

.....

II. State the volume of hydrogen collected at 25 s.

[1]

volume = cm³

III. At what time did the reaction stop?

[1]

time = s

(b) (i) State the purpose of the catalyst.

[1]

.....

(ii) The mass of catalyst added at the start of the experiment was 1.5 g.

State the mass of catalyst remaining after 60 s.

[1]

mass = g

(iii) State how the **volumes in the table** would be different if the catalyst had not been added.

[1]

.....

(c) (i) State how the **volumes in the table** would be affected by using the same mass of zinc **powder** instead of large pieces of zinc.

[1]

.....

(ii) Explain how the **volumes in the table** would be affected if the original experiment was repeated at a higher temperature.

[2]

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.....

.....



6. (a) State, in terms of electrons, the difference between ionic and covalent bonding. [2]

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- (b) When magnesium reacts with oxygen, the ionic compound magnesium oxide is formed.

- (i) The outer electrons of magnesium and oxygen are shown below. Draw arrows to show the transfer of electrons when magnesium oxide is formed. [1]



- (ii) State the charges on the ions formed [2]

Mg O



7. Inos is a Welsh caravan manufacturer based in Denbighshire. The materials used to build modern caravans are very different to those used in early caravans.

Table 1 shows how the materials used in manufacturing the body of a caravan have changed over time.

Year	Materials used
1920	wood
1950	steel and wood
1970	aluminium and wood
2015	glass reinforced plastic (GRP), polyester

Table 2 shows information about some of the materials used to make the body of caravans.

Material		Density (g/cm ³)	Stiffness (GPa)	Melting Point (°C)	Strength (MPa)	Is it brittle?	Does it corrode?
wood		0.6			1 000	No	no but rots when wet
steel		7.8	210	1 357	1 200	No	yes producing rust
aluminium		2.7	69	660	90	No	yes producing a resistant coating
GRP	glass fibres	2.4	180	1 400	3 500	Yes	No
	epoxy resin	1.5			60	Yes	No
polyester		1.9	150	121	250	Yes	No

Use the information above to answer the following questions.

- (a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight:

.....

Strength:

.....



- (b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1.

2.

- (ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

.....

- (c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

- (i) A GRP panel contains a volume of 300 cm^3 of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

mass = g

- (ii) I. State the strength of glass fibre in N/cm^2 . [2]

$$(1 \text{ MPa} = 100 \text{ N/cm}^2)$$

strength = N/cm^2

- II. The cross-sectional area (csa) of a glass fibre is 0.00015 cm^2 .

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = N

END OF PAPER





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1 4 He Helium 2																																																																							
7	Li	3	Lithium	9	Be	4	Beryllium	11	Na	12	Magnesium	13	Al	14	Si	15	Phosphorus	16	S	17	Cl	18	Chlorine	19	F	20	Ne	10	Neon																																										
23	Na	11	Sodium	24	Mg	12	Magnesium	27	Al	13	Aluminium	28	Si	14	Silicon	31	P	15	Phosphorus	32	S	16	Sulfur	35.5	Cl	17	Chlorine	40	Ar	18	Argon																																								
39	K	19	Potassium	40	Ca	20	Calcium	45	Sc	21	Scandium	48	Ti	22	Titanium	51	V	23	Vanadium	52	Cr	24	Chromium	55	Mn	25	Manganese	56	Fe	26	Iron	59	Co	27	Cobalt	59	Ni	28	Nickel	63.5	Cu	29	Copper	65	Zn	30	Zinc	70	Ga	31	Gallium	73	Ge	32	Germanium	75	As	33	Arsenic	79	Se	34	Selenium	80	Br	35	Bromine	84	Kr	36	Krypton
86	Rb	37	Rubidium	88	Sr	38	Strontium	89	Y	39	Yttrium	91	Zr	40	Zirconium	93	Nb	41	Niobium	96	Mo	42	Molybdenum	99	Tc	43	Technetium	101	Ru	44	Ruthenium	103	Rh	45	Rhodium	106	Pd	46	Palladium	108	Ag	47	Silver	112	Cd	48	Cadmium	115	In	49	Indium	119	Sn	50	Tin	122	Sb	51	Antimony	127	I	53	Iodine	131	Xe	54	Xenon				
133	Cs	55	Caesium	137	Ba	56	Barium	139	La	57	Lanthanum	179	Hf	72	Hafnium	181	Ta	73	Tantalum	184	W	74	Tungsten	186	Re	75	Rhenium	190	Os	76	Osmium	192	Ir	77	Iridium	195	Pt	78	Platinum	197	Au	79	Gold	201	Hg	80	Mercury	204	Tl	81	Thallium	207	Pb	82	Lead	209	Bi	83	Bismuth	210	Po	84	Polonium	210	At	85	Astatine	222	Rn	86	Radon
223	Fr	87	Francium	226	Ra	88	Radium	227	Ac	89	Actinium																																																												

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number